Approved by Associate Dean:

06 January 2026

Signature

GAME3015 – Game Engine Development II

Course Description

This senior level project-based course will allow the students to develop a complete 3D game using an industry-standard game development platform. The purpose of this course is to give the students hands-on experience with all aspects of video game development and delivery, as well as working in a team environment. It will bring together topics such as level design, model design and animation, sounds, events, artificial intelligence, and user interface design.

Course Outcomes

1. Explain the principles that make 3D graphics and scenes effective.
2. Implement collision and physics into a game engine.
3. Implement AI and networking into a game engine.
4. Develop a game using a custom engine.
5. Develop a game editor in order to quickly and effectively modify a game.
6. Use advanced debugging and profiling techniques in game creation.
7. Format all deliverables to comply with Canadian laws and policies.

List of Textbooks and Other Teaching Aids:

Required:

Game Engine Architecture, Third Edition
By: Jason Gregory
ISBN-13: 978-1-1380-3545-4
Publisher: CRC Press Taylor & Francis Group

Introduction to 3D Game Programming with DirectX12
By: Frank D. Luna
ISBN: 9781942270065
Publisher: Mercury

Recommended Resources:

None

Course Delivery Mode

- This course has a separate 3-hour session for each section per week.
- **T163:** All classes are on-campus.

  Any variation to the above will be posted in the online course shell in advance.

Assignment Policy

All assignments must be submitted on the due date of each respective assignment by means specified by their professor for that assignment. For every day past the due date there will be 10% penalty unless the student has notified the professor (via e-mail, phone or in person) ahead of due date that he/she has a valid reason for late submission. Submissions will no longer be accepted after five days past an assignment due date.

Test Policy

Students are required to complete lab tests, quizzes, exams as well as take-home assignments. If a student misses a test for valid reasons, including medical, and can provide a doctor's note, he/she will be given a chance to rewrite the test at a later date.

Students are required to adhere to all George Brown College policies and procedures regarding withdrawals, exemptions, attendance, class assignments and academic dishonesty. Please refer to the following: https://www.georgebrown.ca/about/policies/.

In-Person Exam Policy

Mid-term and Final exams for the T163 programs will be conducted in person. Please note the following exam schedule:

- Mid-Term Exams (If applicable): Week 7 of the semester
- Final Exams (If applicable): Week 15 of the semester

Students are expected to be available in person during these exam periods.

Important Note on the Use of Generative AI:

Students must review the "Generative AI Usage Guidelines" document, available on D2L, for detailed instructions on how generative AI tools may be used in this course. The course evaluation table now includes a column labelled "AI Usage Allowed," indicating whether AI use is permitted for each assessment.

Yes: AI can be used with proper referencing.
No: AI cannot be used, and any usage will be considered plagiarism and subject to academic penalties.

Misuse of AI in assessments where it is not permitted or failure to adequately disclose its use will be treated as a violation of academic integrity. According to college policy, consequences may include failing the assignment or the course or more severe disciplinary actions. **Students must also download the AI Usage Declaration form from D2L, complete it, and submit it with their assignments where AI use is permitted.** Adherence to these guidelines is mandatory to maintain academic integrity.

Detailed Evaluation System

Assessment Tool:	Description:	Outcome(s) assessed:	EES assessed:	Date / Week:	% of Final Grade:
Assignment 1	Practical game engine coding assignment	1-7	4-6	5	10%
Assignment 2	Practical game engine coding assignment	1-7	4-6	13	10%
Project	Game engine project - Milestone 1: Prototype - Milestone 2: Demo	1-7	4-6	 7 15	 25% 25%
Final Exam	Written test on code and theory	1-7	4-6	15	30%
		TOTAL:			**100%**

Topical Outline

Learning Schedule / Topical Outline (subject to change with notification)

Week	Topic / Task	Outcomes	Content / Activities	Resources
1	- Animation Systems	1,2	Lab	GEA-Ch. 12 IGP-Ch. 22
2	- Environment Map	1,2	Lab	IGP-Ch. 18
3	- Dynamic Cube map	1,2	Lab	IGP-Ch. 19
4	- Normal Mapping	1,3	Lab	IGP-Ch. 19
5	- Shadow Mapping **- Assignment 1 Due**	1,3, 7	Lab	IGP-Ch. 20
6	- Building Game States	4	Lab	GEA-Ch. 15
7	- Ambient Occlusion **- Project Milestone 1 Due**	4, 7	Lab	IGP-Ch. 21
8	*INTERSESSION WEEK*			
9	- Skeleton Animation	5	Lab	IGP-Ch. 23
10	- AI, Projectiles, and Collision	6	Lab	GEA-Ch. 13
11	- Particle Systems	4, 6	Lab	GEA-Ch. 24
12	- Audio Implementation	1-6	Review	GEA-Ch. 15
13	- Multiplayer Systems **- Assignment 2 Due**	1-6, 7	Review	GEA-Ch. 16
14	- Work Period	1-6	Review	
15	**- Project Milestone 2 Due** **FINAL EXAM**	1-6, 7		

Please note: this schedule may change as resources and circumstances require.
For information on withdrawing from this course without academic penalty, please refer to the College Academic Calendar: **http://www.georgebrown.ca/Admin/Registr/PSCal.aspx**